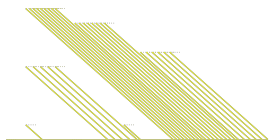


Chapitre 1 : Le langage R

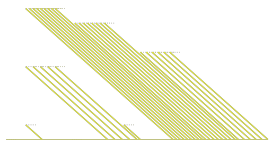
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1 Avant de commencer

Ce chapitre reprends les codes du chapitre 1 sur **le langage R** du livre **Langage R et Statistiques** : Initiation à l'analyse de données publié par les éditions ENI (2^{ème} édition).

```
tibble::view(women)
```

2 Installation de R et RStudio

2.1 installation de l'addin {esquisse}

```
install.packages("esquisse")
```

2.2 installation du {tidyverse}

```
install.packages("tidyverse")
```

3 Fonctions et package

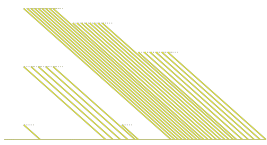
3.1 utilisation du préfixe

```
# ne fonctionne pas
data.table(
  x = seq(from = 1, to = 50, by = 5)
)
```

```
Error in `data.table()` :
! impossible de trouver la fonction "data.table"
```

```
# fonctionne
data.table::data.table(
  x = seq(from = 1, to = 50, by = 5)
)
```

```
      x
<num>
1:    1
2:    6
3:   11
4:   16
```



```
5: 21
6: 26
7: 31
8: 36
9: 41
10: 46
```

3.2 affichage de la page d'aide

```
?seq
```

4 Utilisation du format projet

4.1 Création d'un dossier data

```
dir.create("data")
```

5 Initiation au codage en R

5.1 Nomenclature du langage R

5.1.a Calcul d'une somme

on calcule la somme de 3+8

```
3 + 8
```

```
[1] 11
```

5.1.b Création d'un objet

```
ceci_est_un_objet <- 4.5 + 1.1
```

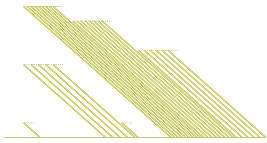
afficher un objet dans la console en utilisant son nom

```
ceci_est_un_objet
```

```
[1] 5.6
```

il est possible de faire de l'utiliser

```
ceci_est_un_objet * 2
```



```
[1] 11.2
```

mais cela ne modifie pas l'objet

```
ceci_est_un_objet
```

```
[1] 5.6
```

5.1.c Modification d'un objet

```
ceci_est_un_objet <- ceci_est_un_objet * 2
```

```
ceci_est_un_objet
```

```
[1] 11.2
```

5.1.d Création séquence de nombre

Pour afficher la page d'aide, il faut utiliser ?seq

```
seq(from = 1, to = 50, by = 5)
```

```
[1] 1 6 11 16 21 26 31 36 41 46
```

5.2 Structure et type des données

5.2.a Nature des données

```
class(iris)
```

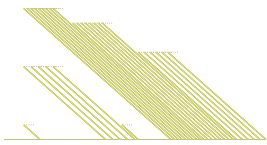
```
[1] "data.frame"
```

```
class(iris$Species)
```

```
[1] "factor"
```

Le signe \$ permet d'accéder à une colonne ou un élément de liste.

```
class(iris$Petal.Width)
```



```
[1] "numeric"
```

5.2.b Structure des données

création d'un vecteur

```
(vecteur <- c(1:10))
```

```
[1] 1 2 3 4 5 6 7 8 9 10
```

création d'une matrice

```
(matrice <- matrix(data = 1:10, ncol = 2))
```

```
      [,1] [,2]
[1,]    1    6
[2,]    2    7
[3,]    3    8
[4,]    4    9
[5,]    5   10
```

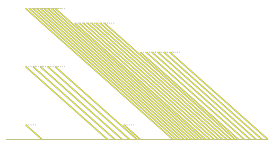
création d'un tableau de données

```
(tableau_de_donnees <- data.frame(vecteur, matrice))
```

```
   vecteur X1 X2
1         1  1  6
2         2  2  7
3         3  3  8
4         4  4  9
5         5  5 10
6         6  1  6
7         7  2  7
8         8  3  8
9         9  4  9
10        10  5 10
```

création d'une liste

```
(liste <- list(vecteur, LETTERS[1:5], tableau_de_donnees))
```



```
[[1]]
[1] 1 2 3 4 5 6 7 8 9 10

[[2]]
[1] "A" "B" "C" "D" "E"

[[3]]
  vecteur X1 X2
1      1 1 6
2      2 2 7
3      3 3 8
4      4 4 9
5      5 5 10
6      6 1 6
7      7 2 7
8      8 3 8
9      9 4 9
10     10 5 10
```

5.3 Importer des données

5.3.a Jeux de données disponibles dans R

```
data()
```

5.3.b Format csv

Création d'un fichier csv

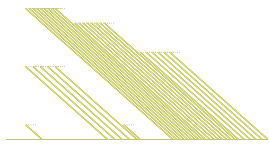
```
write.csv(
  mtcars,
  "data/jdd_voiture.csv"
)
```

lecture d'un fichier csv

```
jdd_voiture <- read.csv(
  file = "data/jdd_voiture.csv",
  row.names = "X"
)
```

visualisation du jeu de données

```
tibble::view(jdd_voiture)
```



création d'un résumé du jeu de données

```
summary(jdd_voiture)
```

mpg	cyl	disp	hp
Min. :10.40	Min. :4.000	Min. : 71.1	Min. : 52.0
1st Qu.:15.43	1st Qu.:4.000	1st Qu.:120.8	1st Qu.: 96.5
Median :19.20	Median :6.000	Median :196.3	Median :123.0
Mean :20.09	Mean :6.188	Mean :230.7	Mean :146.7
3rd Qu.:22.80	3rd Qu.:8.000	3rd Qu.:326.0	3rd Qu.:180.0
Max. :33.90	Max. :8.000	Max. :472.0	Max. :335.0
drat	wt	qsec	vs
Min. :2.760	Min. :1.513	Min. :14.50	Min. :0.0000
1st Qu.:3.080	1st Qu.:2.581	1st Qu.:16.89	1st Qu.:0.0000
Median :3.695	Median :3.325	Median :17.71	Median :0.0000
Mean :3.597	Mean :3.217	Mean :17.85	Mean :0.4375
3rd Qu.:3.920	3rd Qu.:3.610	3rd Qu.:18.90	3rd Qu.:1.0000
Max. :4.930	Max. :5.424	Max. :22.90	Max. :1.0000
am	gear	carb	
Min. :0.0000	Min. :3.000	Min. :1.000	
1st Qu.:0.0000	1st Qu.:3.000	1st Qu.:2.000	
Median :0.0000	Median :4.000	Median :2.000	
Mean :0.4062	Mean :3.688	Mean :2.812	
3rd Qu.:1.0000	3rd Qu.:4.000	3rd Qu.:4.000	
Max. :1.0000	Max. :5.000	Max. :8.000	

5.3.c Format **xlsx**

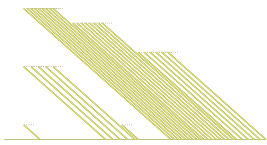
creation d'un fichier excel

```
library(openxlsx)

write.xlsx(
  women,
  "data/jdd_femme.xlsx"
)
```

lire un fichier excel

```
jdd_femme <- read.xlsx(
  "data/jdd_femme.xlsx"
)
```

5.3.d Format txt

Lecture de données depuis un site internet

```
library(rvest)
page <- read_html("https://fr.m.wikisource.org/wiki/Lettres_de_la_comtesse_de_
S%C3%A9gur/Texte_entier") |>
  html_nodes("h1") |>
  html_text()
```

Enregistrement dans un fichier txt

```
write(page, "data/titre_segur.txt")
# ou
page |>
  write("data/titre_segur.txt")
```

lecture d'un fichier txt

```
titre_segur <- read.delim(
  "data/titre_segur.txt",
  header = FALSE
)
```

5.4 utilisation du pipe natif

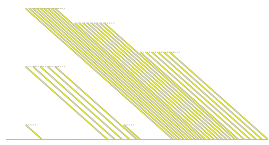
sans le pipe, en créant des objets

```
# création d'objets intermédiaire
etape1 <- c(1:30)
etape2 <- sqrt(etape1)
etape3 <- mean(etape2)
floor(etape3)
```

```
[1] 3
```

```
# sans pipe
floor(mean(sqrt(c(1:10)*3)))
```

```
[1] 3
```



```
# équivalent à  
floor(  
  mean(  
    sqrt(  
      c(1:30)  
    )  
  )  
)
```

```
[1] 3
```

```
# avec pipe  
c(1:30) |>  
  sqrt() |>  
  mean() |>  
  floor()
```

```
[1] 3
```

6 Travailler sur des jeux de données

6.1 Manipulation des données dans R

6.1.a Sélection ligne(s) et/ou colonne(s)

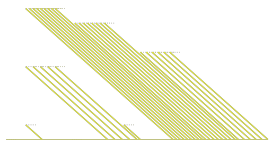
langage de base

```
(exemple <- matrix(data = c(12, 53, 6, 8, 97, 42), nrow = 3))
```

```
      [,1] [,2]  
[1,]   12    8  
[2,]   53   97  
[3,]    6   42
```

```
exemple[3,2]
```

```
[1] 42
```



```
colnames(exemple) <- list("premiere_colonne", "deuxieme_colonne")
rownames(exemple) <- c("premiere_ligne", "deuxieme_ligne", "troisieme_ligne")
exemple
```

	premiere_colonne	deuxieme_colonne
premiere_ligne	12	8
deuxieme_ligne	53	97
troisieme_ligne	6	42

```
exemple["troisieme_ligne", "deuxieme_colonne"]
```

```
[1] 42
```

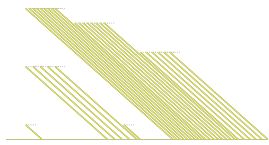
exemple en {tidyverse}

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2
— Conflicts — tidyverse_conflicts() —
* dplyr::filter() masks stats::filter()
* readr::guess_encoding() masks rvest::guess_encoding()
* dplyr::lag() masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
(exemple_tibble <- tibble(
  premiere_colonne = c(12, 53, 6),
  deuxieme_colonne = c(8, 97, 42)
))
```

```
# A tibble: 3 × 2
  premiere_colonne deuxieme_colonne
      <dbl>         <dbl>
1         12             8
2         53            97
3           6            42
```



1	12	8
2	53	97
3	6	42

```
exemple_tibble[3,2]
```

```
# A tibble: 1 × 1  
  deuxieme_colonne  
      <dbl>  
1             42
```

```
exemple_tibble[3, "deuxieme_colonne"]
```

```
# A tibble: 1 × 1  
  deuxieme_colonne  
      <dbl>  
1             42
```

6.1.b Sélection de plusieurs lignes ou colonnes

```
exemple[c(1,3),]
```

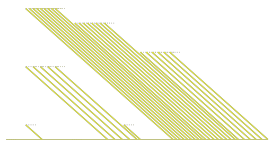
	premiere_colonne	deuxieme_colonne
premiere_ligne	12	8
troisieme_ligne	6	42

```
exemple_tibble[c(1:2), 1]
```

```
# A tibble: 2 × 1  
  premiere_colonne  
      <dbl>  
1             12  
2             53
```

6.1.c Utilisation des parenthèses autour du code

```
test_affichage <- "Ceci est un test pour montrer l'utilisation des parenthèses"  
(test_affichage <- "Ceci est un test pour montrer l'utilisation des parenthèses")
```

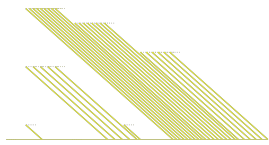


```
[1] "Ceci est un test pour montrer l'utilisation des parenthèses"
```

6.1.d Choix de lignes et colonnes

```
iris[iris$Species == "setosa", ]
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa
18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa
28	5.2	3.5	1.5	0.2	setosa
29	5.2	3.4	1.4	0.2	setosa
30	4.7	3.2	1.6	0.2	setosa
31	4.8	3.1	1.6	0.2	setosa
32	5.4	3.4	1.5	0.4	setosa
33	5.2	4.1	1.5	0.1	setosa
34	5.5	4.2	1.4	0.2	setosa
35	4.9	3.1	1.5	0.2	setosa
36	5.0	3.2	1.2	0.2	setosa



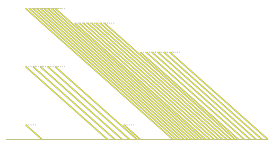
37	5.5	3.5	1.3	0.2	setosa
38	4.9	3.6	1.4	0.1	setosa
39	4.4	3.0	1.3	0.2	setosa
40	5.1	3.4	1.5	0.2	setosa
41	5.0	3.5	1.3	0.3	setosa
42	4.5	2.3	1.3	0.3	setosa
43	4.4	3.2	1.3	0.2	setosa
44	5.0	3.5	1.6	0.6	setosa
45	5.1	3.8	1.9	0.4	setosa
46	4.8	3.0	1.4	0.3	setosa
47	5.1	3.8	1.6	0.2	setosa
48	4.6	3.2	1.4	0.2	setosa
49	5.3	3.7	1.5	0.2	setosa
50	5.0	3.3	1.4	0.2	setosa

```
iris[iris$Species == "setosa", "Sepal.Length"]
```

```
[1] 5.1 4.9 4.7 4.6 5.0 5.4 4.6 5.0 4.4 4.9 5.4 4.8 4.8 4.3 5.8 5.7 5.4 5.1 5.7
[20] 5.1 5.4 5.1 4.6 5.1 4.8 5.0 5.0 5.2 5.2 4.7 4.8 5.4 5.2 5.5 4.9 5.0 5.5 4.9
[39] 4.4 5.1 5.0 4.5 4.4 5.0 5.1 4.8 5.1 4.6 5.3 5.0
```

```
iris[iris$Species == "setosa" | iris$Sepal.Length < 5, ]
```

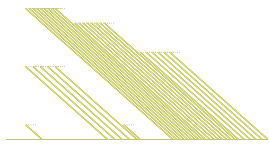
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa



18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa
28	5.2	3.5	1.5	0.2	setosa
29	5.2	3.4	1.4	0.2	setosa
30	4.7	3.2	1.6	0.2	setosa
31	4.8	3.1	1.6	0.2	setosa
32	5.4	3.4	1.5	0.4	setosa
33	5.2	4.1	1.5	0.1	setosa
34	5.5	4.2	1.4	0.2	setosa
35	4.9	3.1	1.5	0.2	setosa
36	5.0	3.2	1.2	0.2	setosa
37	5.5	3.5	1.3	0.2	setosa
38	4.9	3.6	1.4	0.1	setosa
39	4.4	3.0	1.3	0.2	setosa
40	5.1	3.4	1.5	0.2	setosa
41	5.0	3.5	1.3	0.3	setosa
42	4.5	2.3	1.3	0.3	setosa
43	4.4	3.2	1.3	0.2	setosa
44	5.0	3.5	1.6	0.6	setosa
45	5.1	3.8	1.9	0.4	setosa
46	4.8	3.0	1.4	0.3	setosa
47	5.1	3.8	1.6	0.2	setosa
48	4.6	3.2	1.4	0.2	setosa
49	5.3	3.7	1.5	0.2	setosa
50	5.0	3.3	1.4	0.2	setosa
58	4.9	2.4	3.3	1.0	versicolor
107	4.9	2.5	4.5	1.7	virginica

```
iris[iris$Species == "setosa" & iris$Sepal.Length < 5, ]
```

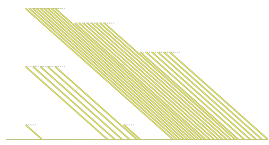
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa



7	4.6	3.4	1.4	0.3	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
23	4.6	3.6	1.0	0.2	setosa
25	4.8	3.4	1.9	0.2	setosa
30	4.7	3.2	1.6	0.2	setosa
31	4.8	3.1	1.6	0.2	setosa
35	4.9	3.1	1.5	0.2	setosa
38	4.9	3.6	1.4	0.1	setosa
39	4.4	3.0	1.3	0.2	setosa
42	4.5	2.3	1.3	0.3	setosa
43	4.4	3.2	1.3	0.2	setosa
46	4.8	3.0	1.4	0.3	setosa
48	4.6	3.2	1.4	0.2	setosa

```
filter(iris, Species == "setosa")
```

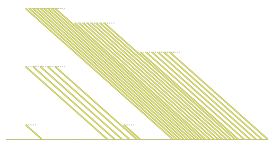
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa
18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa



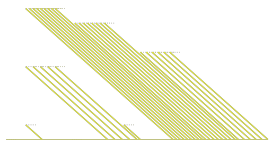
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa
28	5.2	3.5	1.5	0.2	setosa
29	5.2	3.4	1.4	0.2	setosa
30	4.7	3.2	1.6	0.2	setosa
31	4.8	3.1	1.6	0.2	setosa
32	5.4	3.4	1.5	0.4	setosa
33	5.2	4.1	1.5	0.1	setosa
34	5.5	4.2	1.4	0.2	setosa
35	4.9	3.1	1.5	0.2	setosa
36	5.0	3.2	1.2	0.2	setosa
37	5.5	3.5	1.3	0.2	setosa
38	4.9	3.6	1.4	0.1	setosa
39	4.4	3.0	1.3	0.2	setosa
40	5.1	3.4	1.5	0.2	setosa
41	5.0	3.5	1.3	0.3	setosa
42	4.5	2.3	1.3	0.3	setosa
43	4.4	3.2	1.3	0.2	setosa
44	5.0	3.5	1.6	0.6	setosa
45	5.1	3.8	1.9	0.4	setosa
46	4.8	3.0	1.4	0.3	setosa
47	5.1	3.8	1.6	0.2	setosa
48	4.6	3.2	1.4	0.2	setosa
49	5.3	3.7	1.5	0.2	setosa
50	5.0	3.3	1.4	0.2	setosa

```
filter(iris, Species == "setosa") |> select(Sepal.Length)
```

	Sepal.Length
1	5.1
2	4.9
3	4.7
4	4.6
5	5.0
6	5.4
7	4.6
8	5.0
9	4.4

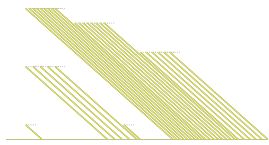


10	4.9
11	5.4
12	4.8
13	4.8
14	4.3
15	5.8
16	5.7
17	5.4
18	5.1
19	5.7
20	5.1
21	5.4
22	5.1
23	4.6
24	5.1
25	4.8
26	5.0
27	5.0
28	5.2
29	5.2
30	4.7
31	4.8
32	5.4
33	5.2
34	5.5
35	4.9
36	5.0
37	5.5
38	4.9
39	4.4
40	5.1
41	5.0
42	4.5
43	4.4
44	5.0
45	5.1
46	4.8
47	5.1
48	4.6
49	5.3
50	5.0



```
filter(iris, Species == "setosa" | iris$Sepal.Length < 5)
```

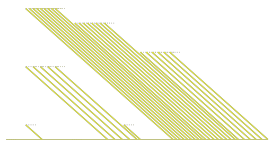
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa
18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa
28	5.2	3.5	1.5	0.2	setosa
29	5.2	3.4	1.4	0.2	setosa
30	4.7	3.2	1.6	0.2	setosa
31	4.8	3.1	1.6	0.2	setosa
32	5.4	3.4	1.5	0.4	setosa
33	5.2	4.1	1.5	0.1	setosa
34	5.5	4.2	1.4	0.2	setosa
35	4.9	3.1	1.5	0.2	setosa
36	5.0	3.2	1.2	0.2	setosa
37	5.5	3.5	1.3	0.2	setosa
38	4.9	3.6	1.4	0.1	setosa
39	4.4	3.0	1.3	0.2	setosa



40	5.1	3.4	1.5	0.2	setosa
41	5.0	3.5	1.3	0.3	setosa
42	4.5	2.3	1.3	0.3	setosa
43	4.4	3.2	1.3	0.2	setosa
44	5.0	3.5	1.6	0.6	setosa
45	5.1	3.8	1.9	0.4	setosa
46	4.8	3.0	1.4	0.3	setosa
47	5.1	3.8	1.6	0.2	setosa
48	4.6	3.2	1.4	0.2	setosa
49	5.3	3.7	1.5	0.2	setosa
50	5.0	3.3	1.4	0.2	setosa
51	4.9	2.4	3.3	1.0	versicolor
52	4.9	2.5	4.5	1.7	virginica

```
filter(iris, Species == "setosa" & iris$Sepal.Length < 5)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	4.6	3.4	1.4	0.3	setosa
5	4.4	2.9	1.4	0.2	setosa
6	4.9	3.1	1.5	0.1	setosa
7	4.8	3.4	1.6	0.2	setosa
8	4.8	3.0	1.4	0.1	setosa
9	4.3	3.0	1.1	0.1	setosa
10	4.6	3.6	1.0	0.2	setosa
11	4.8	3.4	1.9	0.2	setosa
12	4.7	3.2	1.6	0.2	setosa
13	4.8	3.1	1.6	0.2	setosa
14	4.9	3.1	1.5	0.2	setosa
15	4.9	3.6	1.4	0.1	setosa
16	4.4	3.0	1.3	0.2	setosa
17	4.5	2.3	1.3	0.3	setosa
18	4.4	3.2	1.3	0.2	setosa
19	4.8	3.0	1.4	0.3	setosa
20	4.6	3.2	1.4	0.2	setosa



6.2 Jointure

6.2.a création des tables

```
(table_gauche <- tibble(  
  id = LETTERS[1:3],  
  x = c(5, 9, 7),  
  y = c(1, 4, 8)  
))
```

```
# A tibble: 3 × 3  
  id      x      y  
  <chr> <dbl> <dbl>  
1 A      5      1  
2 B      9      4  
3 C      7      8
```

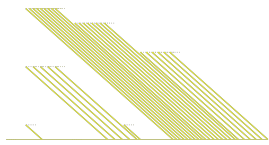
```
(table_droite <- tibble(  
  identite = LETTERS[c(1, 2, 4)],  
  z = c(4, 7, 6),  
  w = c(2, 8, 6)  
))
```

```
# A tibble: 3 × 3  
  identite      z      w  
  <chr>    <dbl> <dbl>  
1 A      4      2  
2 B      7      8  
3 D      6      6
```

6.2.b jointures

```
inner_join(  
  table_gauche,  
  table_droite,  
  by = c("id" = "identite")  
)
```

```
# A tibble: 2 × 5  
  id      x      y      z      w  
  <chr> <dbl> <dbl> <dbl> <dbl>
```



1	A	5	1	4	2
2	B	9	4	7	8

```
full_join(  
  table_gauche,  
  table_droite,  
  by = c("id" = "identite")  
)
```

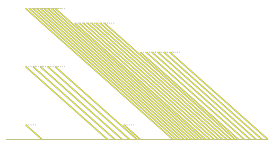
```
# A tibble: 4 × 5  
  id      x      y      z      w  
  <chr> <dbl> <dbl> <dbl> <dbl>  
1 A      5      1      4      2  
2 B      9      4      7      8  
3 C      7      8     NA     NA  
4 D     NA     NA      6      6
```

```
left_join(  
  table_gauche,  
  table_droite,  
  by = c("id" = "identite")  
)
```

```
# A tibble: 3 × 5  
  id      x      y      z      w  
  <chr> <dbl> <dbl> <dbl> <dbl>  
1 A      5      1      4      2  
2 B      9      4      7      8  
3 C      7      8     NA     NA
```

```
right_join(  
  table_gauche,  
  table_droite,  
  by = c("id" = "identite")  
)
```

```
# A tibble: 3 × 5  
  id      x      y      z      w  
  <chr> <dbl> <dbl> <dbl> <dbl>  
1 A      5      1      4      2
```



2 B	9	4	7	8
3 D	NA	NA	6	6

6.2.c vérification

```
summary(table_gauche)
```

	id	x	y
Length	:3	Min. :5	Min. :1.000
N.unique	:3	1st Qu.:6	1st Qu.:2.500
N.blank	:0	Median :7	Median :4.000
Min.nchar	:1	Mean :7	Mean :4.333
Max.nchar	:1	3rd Qu.:8	3rd Qu.:6.000
		Max. :9	Max. :8.000

```
summary(table_droite)
```

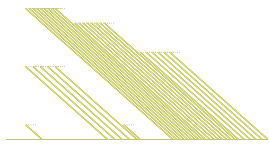
	identite	z	w
Length	:3	Min. :4.000	Min. :2.000
N.unique	:3	1st Qu.:5.000	1st Qu.:4.000
N.blank	:0	Median :6.000	Median :6.000
Min.nchar	:1	Mean :5.667	Mean :5.333
Max.nchar	:1	3rd Qu.:6.500	3rd Qu.:7.000
		Max. :7.000	Max. :8.000

```
table_gauche |>  
  count(id) |>  
  filter(n > 1)
```

```
# A tibble: 0 × 2  
# i 2 variables: id <chr>, n <int>
```

```
table_droite |>  
  count(identite) |>  
  filter(n > 1)
```

```
# A tibble: 0 × 2  
# i 2 variables: identite <chr>, n <int>
```



```
table_gauche$id |> class()
```

```
[1] "character"
```

```
table_droite$identite |> class()
```

```
[1] "character"
```

```
nrow(table_gauche)
```

```
[1] 3
```

```
nrow(table_droite)
```

```
[1] 3
```

```
table_gauche$id %in% table_droite$identite |> sum()
```

```
[1] 2
```